Linear Regression and OLS

ANSWER KEY

Section 1: Simple Linear Regression and Residuals

1 In simple linear regression $\hat{y} = 3.5x + 12$, what is the residual $e_i = y_i - \hat{y}_i$ for an observation at $(x_i, y_i) = (4, 30)$?

a. $+4$ ✓

b. -4

c. $+2$

d. -2

$\hat{y} = 3.5(4) + 12 = 14 + 12 = 26$. Residual $e = 30 - 26 = +4$.

2 What is the mathematical condition guaranteed by the OLS fit regarding the sum of all residuals $\sum_{i=1}^{n} e_i$?

a. $\sum e_i = 0$ **exactly** ✓

b. $\sum e_i = 1$

c. $\sum e_i = n$

d. $\sum e_i = \text{SSE}$

The first normal equation $\frac{\partial \text{SSE}}{\partial \beta_0} = -2\sum(y_i - \beta_0 - \beta_1 x_i) = 0$ guarantees $\sum e_i = 0$.

3 Given $\bar{x} = 10$, $\bar{y} = 25$, and fitted slope $\hat{\beta}_1 = 2$, compute the y-intercept $\hat{\beta}_0$.

a. 5 ✓

b. 15

c. 20

d. 10

Regression line passes through the centroid: $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 25 - 2(10) = 5$.

Section 2: Analytical OLS Formulas and Covariance

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 For a dataset with $\text{Cov}(x, y) = 18$ and $\text{Var}(x) = 6$, compute the OLS slope $\hat{\beta}_1$.

a. 3.0 ✓

b. 0.33

c. 12

d. 108

$\hat{\beta}_1 = \frac{\text{Cov}(x,y)}{\text{Var}(x)} = \frac{18}{6} = 3.0.$

5 If the Pearson correlation coefficient between x and y is $r = 0.8$, with $\sigma_x = 2$ and $\sigma_y = 5$, find the slope $\hat{\beta}_1$.

a. 2.0 ✓

b. 1.25

c. 0.32

d. 4.0

$\hat{\beta}_1 = r\frac{\sigma_y}{\sigma_x} = 0.8\left(\frac{5}{2}\right) = 0.8(2.5) = 2.0.$

6 If all x values in a dataset are scaled by a factor of 10, how does the OLS regression slope change?

a. **It is divided by** 10 ✓

b. It is multiplied by 10

c. It is multiplied by 100

d. It remains unchanged

$\beta_1' = \frac{\text{Cov}(10x,y)}{\text{Var}(10x)} = \frac{10\text{Cov}(x,y)}{100\text{Var}(x)} = \frac{1}{10}\beta_1.$

Section 3: Matrix Normal Equations and Projections

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 State the matrix Normal Equations for the OLS weight vector $\boldsymbol{\beta}$ given design matrix $\mathbf{X}$ and target $\mathbf{y}$.

a. $(\mathbf{X}^T\mathbf{X})\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$ ✓

b. $\mathbf{X}\boldsymbol{\beta} = \mathbf{y}$

c. $(\mathbf{X}\mathbf{X}^T)\boldsymbol{\beta} = \mathbf{y}$

d. $\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$

Minimizing $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 \Rightarrow -2\mathbf{X}^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = 0 \Rightarrow (\mathbf{X}^T\mathbf{X})\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$.

8 When is the Gram matrix $(\mathbf{X}^T\mathbf{X})$ strictly invertible?

a. **When the columns of $\mathbf{X}$ are linearly independent (full column rank)** ✓

b. When $\mathbf{X}$ is a square matrix

c. When all elements in $\mathbf{X}$ are positive

d. When the target $\mathbf{y}$ has zero mean

Full column rank ensures $\mathbf{X}^T\mathbf{X}$ is positive definite and thus invertible.

9 What geometric role does the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ play?

a. **It orthogonal-projects target $\mathbf{y}$ onto the column space of $\mathbf{X}$ ($\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$)** ✓

b. It inverts the covariance matrix

c. It computes feature cross-correlations

d. It normalizes features to zero mean

$\mathbf{H}$ satisfies $\mathbf{H}^2 = \mathbf{H}$ (idempotent) and $\mathbf{H}^T = \mathbf{H}$ (symmetric), the definition of orthogonal projection.

Section 4: Goodness of Fit and Regularization

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 In an evaluation, $\text{SSE} = 25$ and $\text{SST} = 100$. Compute the coefficient of determination R^2.

a. 0.75 ✓

b. 0.25

c. 0.50

d. 0.80

$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{25}{100} = 0.75.$

11 How does Ridge regression (L_2 penalty) resolve multicollinearity when $\mathbf{X}^T\mathbf{X}$ is ill-conditioned?

a. **It adds $\lambda\mathbf{I}$ to $\mathbf{X}^T\mathbf{X}$, guaranteeing invertibility: $(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}$** ✓

b. It drops half the features automatically

c. It sets all correlated weights to zero

d. It eliminates the bias term

Adding $\lambda > 0$ along the diagonal shifts all eigenvalues up by λ, making the matrix well-conditioned.

12 If $R^2 = 0.64$ in simple linear regression, what is the magnitude of the Pearson correlation coefficient $|r|$?

a. 0.80 ✓

b. 0.64

c. 0.32

d. 0.40

In simple bivariate regression, $R^2 = r^2 \Rightarrow |r| = \sqrt{0.64} = 0.80.$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647